Process Characterization Report

A-Mab Drug Substance — Small-Virus Retentive Filtration (Step 9) — PCR-009

Process Development / Manufacturing Science & Technology (MSAT)

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Abbreviations

Table 2: Abbreviations used in this report.

Abbreviation	Expansion
AMV	Analytical method validation
CPP	Critical process parameter
CQA	Critical quality attribute
Cpk	Process capability index (one-sided)
%CV	Coefficient of variation
DoE	Design of experiments
ICH	International Council for Harmonisation
LRF	Log reduction factor
MSAT	Manufacturing Science and Technology
MVM	Minute virus of mice (parvovirus model)
NOR	Normal operating range
PAR	Proven acceptable range
PRESS	Predicted residual error sum of squares
qPCR	Quantitative polymerase chain reaction
SOP	Standard operating procedure
TCID50	Median tissue culture infectious dose
WC-CPP	Well controlled critical process parameter
XMuLV	Xenotropic murine leukaemia virus (enveloped model)

Executive summary

Small-virus retentive filtration is the dedicated small virus removal step of the A-Mab drug substance process. The step removes virus by size on a membrane whose pore size distribution retains the parvoviruses while the antibody passes into the filtrate. It is the last purification step before formulation, and it is the largest single contributor to the parvovirus clearance claimed for the process. This report presents the characterization study executed against the protocol PCP-009 on a qualified small scale spiking model, at conditions equivalent to the commercial scale of 15,000 L. The study follows the lifecycle approach of the FDA process validation guidance (U.S. Food and Drug Administration 2011) and the viral safety framework of ICH Q5A(R2) (International Council for Harmonisation 2023a).

The step forms no quality attribute of its own. It contributes log reduction to two cumulative viral safety attributes, MVM clearance and XMuLV clearance, which are set by the anion exchange step (PCR-008) and the low pH inactivation step (PCR-006). The step carries 2 process parameters, the filtration volume and the filtration pressure, and both were characterized in a multivariate design.

The filtration volume is the only parameter with a demonstrated effect on virus retention. Across the characterized range of 50 to 140 L/m2 the predicted MVM LRF falls by 1.76 log at the set-point pressure, and the effect is significant in the screening design ($p = 0.027$). The response-surface model for MVM LRF is adequate, with an R^2 of 0.947, an adjusted R^2 of 0.903 and a model p-value of 0.0009, and its lack of fit is not significant against the pure error of the centre points. Neither XMuLV LRF nor step yield has an active factor over the ranges studied. The enveloped retrovirus is far larger than the pores of the membrane and is retained across the whole characterized region.

The proven acceptable range for the filtration volume ends at 97.2 L/m2 when the filtration pressure varies within its normal operating range. That bound lies below the routine upper limit of 105 L/m2, so the maximum load is carried into the control strategy at the proven acceptable range rather than at the edge of the normal operating range (§10). Commercial-scale capability was estimated from 2,000 simulated batches. The cumulative MVM clearance carries a one-sided Cpk of 1.51 against a requirement of 8.6 log, and it is the tightest capability in the drug substance register. Both parameters were classified as well controlled critical process parameters. During execution 2 deviations were recorded, and each was dispositioned as retained after impact assessment (§13). The outcome of the study rolls up into PCMR-001.

1 Introduction

1.1 Product and unit operation

A-Mab is a humanized IgG1 monoclonal antibody whose primary mechanism of action is antibody dependent cellular cytotoxicity. It is produced by fed-batch cell culture at a commercial scale of 15,000 L and purified on the Novacyte platform train. The train runs Protein A capture, low pH viral inactivation, cation exchange chromatography, anion exchange chromatography, small-virus retentive filtration and ultrafiltration with diafiltration. Small-virus retentive filtration is Step 9. It receives the anion exchange flow-through pool and delivers the filtrate to the formulation step.

The step removes virus by size. The membrane has a pore size distribution chosen to retain the smallest viruses of concern, the parvoviruses, while the antibody passes freely into the filtrate. The claim made for the step is a log reduction factor for two model viruses, the parvovirus MVM and the enveloped retrovirus XMuLV. The step forms and clears no product quality attribute. The two attributes it governs are cumulative across the process, and this step contributes 5.32 log of the MVM clearance and 5.82 log of the XMuLV clearance credited to the process. Viral safety is claimed modularly from three orthogonal steps, and no claim is made here for the low pH hold or for the anion exchange step, whose contributions are reported in PCR-006 and PCR-008.

1.2 Objectives

The study had five objectives.

(i) Quantify the effect of the filtration volume and the filtration pressure on the retention of MVM and XMuLV and on step yield.

(ii) Define the region of those two parameters over which the step meets the log reduction credited to it.

(iii) Establish a proven acceptable range for each parameter against each attribute the step governs, and locate the normal operating range within it.

(iv) Classify each parameter according to SOP-4001, from the demonstrated effects and the control capability of the commercial equipment.

(v) Justify the ranges carried into Stage 2 and state the step's contribution to the control strategy.

1.3 Regulatory and scientific basis

The study is a Stage 1 process design activity under the lifecycle approach to process validation (U.S. Food and Drug Administration 2011). It follows the enhanced development approach of ICH Q8 (International Council for Harmonisation 2009), ICH Q9 (International Council for Harmonisation 2023b) and ICH Q11 (International Council for Harmonisation 2012), in which prior knowledge and a risk assessment set the scope of experimentation and the result is a control strategy. Criticality is treated as a continuum rather than as a binary label, so a parameter linked to a quality attribute is classified by the risk that it leaves its acceptable range as well as by the size of its effect (International Council for Harmonisation 2023b).

Viral safety is evaluated under ICH Q5A(R2) (International Council for Harmonisation 2023a). The guidance asks for clearance to be demonstrated on a qualified small scale model with spiked virus, for the model viruses to represent the relevant classes, and for the clearance credited to a step to be supported by the operating conditions under which it was measured. The A-Mab case study is the source of the process model, the quality attribute framework and the risk methodology used across this package (CMC Biotech Working Group 2009).

1.4 Related documents

The documents that scope, plan and consolidate this study are listed in Table 1.1. The risk assessment RA-001 set the study scope, the protocol PCP-009 defined the design and the acceptance criteria before the data existed, and this report is its paired deliverable.

Table 1.1: Documents related to this report.

Document ID	Title	Relationship
PTP-001	Process Transfer Plan (A-Mab Drug Substance)	Parent — transfer scope
RA-001	Pre-Characterization Process Risk Assessment (A-Mab Drug Substance)	Parent — risk basis for study scope
PCMP-001	Process Characterization Master Plan (A-Mab Drug Substance)	Parent — master plan
PCP-009	Process Characterization Plan (Small-Virus Retentive Filtration (Step 9))	Sibling — paired document
PCMR-001	Process Characterization Master Report (A-Mab Drug Substance)	Rolls up into

2 Prior knowledge and quality risk basis

2.1 Platform and prior-product knowledge

Small-virus retentive filtration has been operated at commercial scale for three prior humanized IgG1 antibodies on this platform, X-Mab, Y-Mab and Z-Mab, and through A-Mab clinical manufacture. The filter type, the membrane format and the mode of operation are unchanged from those products. The mechanism of removal is sieving, which depends on the size of the virus relative to the pore size distribution and not on the chemistry of the antibody. Prior knowledge therefore transfers to A-Mab, and the purpose of this study was to confirm and bound the platform behaviour rather than to discover it.

Two features of the mechanism set the expectations the study was designed to test. The first is fouling. The membrane accumulates product and residual impurity as the batch is filtered, the flux redistributes toward the least fouled and largest pores, and small virus can pass through them. Retention of the parvovirus was therefore expected to fall as the volume filtered per unit area rises, and the maximum load was expected to bound the clearance claim. The second is the size difference between the two model viruses. MVM sits close to the pore size cut-off of the membrane, while XMuLV is many times larger than the pores. XMuLV retention was therefore expected to be insensitive to the operating conditions as long as the membrane is intact, which the post-use integrity test confirms for each batch.

Filtration pressure was expected to act in two directions. A higher pressure raises the flux and compresses the fouling layer. A low pressure, or an interruption of flow, allows small virus held in the pore structure to diffuse through the membrane. The claim is therefore made over a pressure range and for uninterrupted operation.

2.2 Quality attributes in scope

The quality attributes this step governs are given in Table 2.1 with their acceptance criteria and their criticality. Both are viral safety attributes and both are classified as very high criticality by Tool #1, which combines the impact of the attribute on patient safety with the uncertainty in that impact (CMC Biotech Working Group 2009). Both acceptance criteria are cumulative over the process and are stated as a minimum log reduction, so neither is met or missed by this step alone.

Table 2.1: Quality attributes governed by the step, from the drug substance register.

CQA	Category	Acceptance	Criticality	Tool #1
Viral clearance — MVM (parvovirus)	viral safety	8.6–20 log10 (cumulative)	VH	20
Viral clearance — XMuLV (enveloped)	viral safety	16.7–30 log10 (cumulative)	VH	20

The step sets no attribute of its own. The register records MVM clearance against the anion exchange step and XMuLV clearance against the low pH inactivation step, which are the steps that set the cumulative claim. Product quality attributes formed upstream, such as the glycan and charge variant attributes and aggregate, are not affected by a size based membrane operated in single pass, and no claim of clearance or modification is made for them here.

2.3 Risk-based prioritization and parameter selection

RA-001 ranked the parameters of every step by their potential impact on a quality attribute and by their potential to interact with other parameters, and assigned each one a study type. For this step the assessment carried 2 parameters, the filtration volume and the filtration pressure, and assigned both to the multivariate design. No parameter of this step required univariate support. The assessment named the volume filtered per unit area as the parameter that bounds the clearance claim, on the mechanism set out in §2.1 and on the experience of the three prior products.

Parameters held to platform specification were not characterized. The membrane lot, the filter format and the pre-use and post-use integrity tests are controlled by SOP-2012 and are identity and quality controlled rather than varied. The load material is the anion exchange flow-through pool, whose composition is characterized in PCR-008 and controlled by that step's own operating ranges.

3 Materials and methods

3.1 Scale-down model and its qualification

The study was executed on a small scale spiking model of the commercial filtration step, qualified under SOP-1001. The model uses the same membrane chemistry, the same pore size distribution and the same filter format as the commercial device, in a device of smaller area. The two parameters that define the duty of the membrane were held at commercial equivalents. The volume filtered per unit area is normalized to membrane area, and the filtration pressure is the transmembrane pressure applied across the membrane. Both are scale independent in this sense, which is what allows the model to stand for the commercial device.

Qualification compared the model with at-scale data on the input and output attributes of the step. The load material, the volumetric throughput per unit area, the pressure profile and the step yield were matched, and the filtrate was tested by the same methods used at scale. A spiking model cannot be run at commercial scale, because virus is not introduced into a manufacturing suite. The clearance claim therefore rests on the model, and Q5A(R2) accepts that basis where the model is shown to represent the commercial step (International Council for Harmonisation 2023a). The centre-point replicates of each design provide the pure error term against which lack of fit is tested (§5.1), and they are the direct measure of how reproducible the model is at the conditions the process is run at.

Virus was spiked into the load at a titre high enough to give a measurable reduction across the filter, and the spiked load was clarified before filtration so that the spike itself did not foul the membrane. The log reduction factor is the difference between the virus titre of the load and the titre of the filtrate, on a common volumetric basis. Each run was closed with a post-use filter integrity test under SOP-2012. An integrity test failure invalidates the run, and no run of this study failed it.

3.2 Operation

The step is operated in single pass and in a single filtration cycle per batch. The anion exchange flow-through pool is filtered through a pre-filter and then through the virus retentive membrane, at a controlled transmembrane pressure, until the target volume per unit area is reached. The filtrate is collected in one pool. The membrane is then flushed with buffer to recover product held up in the device, and the flush is included in the pool. Flow is not interrupted during the filtration. The set-point of the filtration volume is 90 L/m2 and the set-point of the filtration pressure is 13 psi.

Two features of the operation are fixed rather than characterized. The filter is used once and discarded, so there is no life-cycle behaviour to characterize as there is for a chromatography resin. The membrane area installed at commercial scale is set from the batch volume and the maximum load, so the load per unit area is the parameter under control and the batch volume is not.

3.3 Analytical methods

The controlled procedures and validated methods that governed the study are listed in Table 3.1. Virus titres were measured by infectivity assay. MVM was titrated by the method validated under AMV-3018 and XMuLV by the method validated under AMV-3017. Both report a titre from which the log reduction factor is calculated. Step yield was determined by mass balance across the filter from the protein concentration and the volume of the load and of the pool. Method performance is summarized in Appendix D.

Table 3.1: Controlled procedures and validated analytical methods for the step.

Reference	Title	Type
SOP-1001	Qualification of Scale-Down Models	SOP
SOP-1002	Design, Execution and Statistical Analysis of DoE Studies	SOP
SOP-2012	Operation and Post-Use Integrity Testing of the Small-Virus Retentive Filtration Step	SOP
SOP-4001	Process-Parameter Classification and Control-Strategy Definition	SOP
AMV-3018	MVM Infectivity Titre (TCID50/qPCR)	Method validation
AMV-3017	XMuLV Infectivity Titre (TCID50)	Method validation

The infectivity assays carry the largest analytical variability of the methods used in this study, and that variability is visible in the centre-point spread reported in §5.1. It is also recognized in the acceptance criterion applied to the step, which sits 0.35 log below the log reduction credited to the step in order to allow for it (§7.1).

3.4 Sampling plan

Each run was sampled twice for virus titre, once from the spiked load before filtration and once from the filtrate pool at the end of the run. Both samples were assayed in the same run of the method, so that the two titres that form a log reduction factor share their assay conditions. Yield samples were drawn from the same two points. The screening design carries 3 centre-point replicates and the response-surface design carries 4, each executed on a separate filter device.

3.5 Statistical methods

Both designs were analysed on coded factors. Each factor is coded so that the midpoint of its characterization range is zero and the edges are minus one and plus one. The screening design was fitted by ordinary least squares with both main effects and their interaction. An effect is reported as twice the coded coefficient, which is the change in the response across the full range of the factor. The response-surface design was fitted with a full quadratic model in the same two factors. Significance is judged at $\alpha = 0.05$ throughout.

Model adequacy is reported as R^2, adjusted R^2, predicted R^2 from the PRESS statistic, and the F test of the model against the residual. Lack of fit is tested by partitioning the residual sum of squares into a lack-of-fit term and the pure error of the centre-point replicates. A response with no active factor is reported as such rather than as a fitted model, because a model with no significant term predicts nothing that the centre-point mean does not.

The screening design identifies which factors act. The response-surface model is the predictive model and is the basis of the design space and of the proven acceptable ranges. Two proven acceptable range analyses were run for each parameter against each attribute the step governs. The first fixes the other parameter and scans the parameter of interest across its characterization range. The second propagates the other parameter across its normal operating range by Monte-Carlo sampling of the fitted model with its residual variance, and requires the 95 % predictive interval to stay within the criterion. Commercial-scale capability was estimated by Monte-Carlo simulation of 2,000 batches with every parameter of the train sampled inside its normal operating range, and is reported as a one-sided Cpk against the lower limit of each attribute.

4 Study design

4.1 Factors, ranges and the knowledge space

The parameters characterized, their set-points, their normal operating ranges and the ranges over which they were studied are given in Table 4.1. The coded letters used in the effect and coefficient tables are given in Table 4.2.

Table 4.1: Parameters characterized at the step, with the final classification from §9.

Parameter	Unit	Set-point	NOR	Char. range	Class	Study
Filtration volume (load)	L/m2	90	50–105	50–140	WC-CPP	multivariate
Filtration pressure	psi	13	10–20	8–30	WC-CPP	multivariate

Table 4.2: Coded factor legend for the screening and response-surface designs.

Code	Factor	Unit	Studied in
A	Filtration volume (load)	L/m2	screening + RSM
B	Filtration pressure	psi	screening + RSM

The characterized region is wider than the normal operating range for both parameters, which is what makes it a knowledge space rather than a confirmation of the routine ranges. The filtration volume was studied from 50 to 140 L/m2 against a normal operating range of 50 to 105 L/m2. The upper edge sits well above the routine maximum load because the mechanism predicts that retention falls with load, and the point at which it falls below the criterion is the quantity the claim needs. The filtration pressure was studied from 8 to 30 psi against a normal operating range of 10 to 20 psi. That range brackets the routine range on both sides, because the mechanism predicts different failure modes at low and at high pressure.

4.2 Screening design

The screening design is a two-level full factorial in both parameters with 3 centre points, 7 runs in total. The factorial part contributes 4 runs at the corners of the characterized region and estimates both main effects and their interaction. The centre points estimate pure error and detect curvature. The design is not able to separate curvature from either factor, which is what the response-surface design was executed to do. The coded matrix and the measured responses are given in Appendix A.

4.3 Response-surface design

The response-surface design is a face-centred central composite design in the same two parameters, 12 runs in total. It carries 4 factorial runs, 4 axial runs at the faces of the characterized region and 4 centre points. The axial runs place the design at the edges of each parameter with the other at its midpoint, which is what supports the quadratic terms. The design estimates both main effects, their interaction and both quadratic terms, and leaves 6 residual degrees of freedom, 3 of which are pure error. The coded matrix and the measured responses are given in Appendix B.

4.4 Univariate assessment

No parameter of this step was assessed univariately. RA-001 carried 2 parameters into the assessment and assigned both to the multivariate design, so there was no secondary parameter whose range needed support from a one factor at a time study. The inputs that are not parameters of the design, such as the membrane lot and the composition of the load, are controlled to specification rather than varied (§2.3).

5 Results

5.1 Centre-point performance and reproducibility

The centre points of the response-surface design were executed at the midpoint of both parameters on 4 separate filter devices. Their mean, standard deviation and coefficient of variation per response are given in Table 5.1. These replicates are the pure error term of every lack-of-fit test reported below.

Table 5.1: Centre-point reproducibility of the response-surface design.

Response	n	Mean	SD	%CV
MVM LRF ($\log_{10}$)	4	5.54	0.116	2.1
XMuLV LRF ($\log_{10}$)	4	6.13	0.176	2.87
Step yield	4	0.983	0.00839	0.854

The least reproducible response is XMuLV LRF ($\log_{10}$), at 2.87 %CV. MVM LRF follows at 2.10 %CV, which is a standard deviation of 0.12 log. Step yield is the most reproducible response at 0.85 %CV. The spread of both virus responses is dominated by the infectivity assays rather than by the filtration, and it is of the order of the analytical allowance built into the step criterion (§7.1).

The centre points of the screening design spread more widely for MVM LRF, at 0.49 log standard deviation over 3 replicates against 0.12 log over 4 in the response-surface design. The difference is within what three and four replicates of an infectivity assay can resolve, and the response-surface estimate is the one used as pure error for the predictive model.

5.2 Screening effects

Model adequacy for the three responses of the screening design is given in Table 5.2, and the effect estimates for MVM LRF in Table 5.3. An effect is the change in the response across the full characterized range of the factor.

Table 5.2: Screening model adequacy for the three responses.

Response	N	R^2	Adj. R^2	Pred. R^2	F	p-value	RMSE
MVM LRF ($\log_{10}$)	7	0.848	0.695	0.769	5.56	0.0962	0.4

Response	N	R^2	Adj. R^2	Pred. R^2	F	p-value	RMSE
XMuLV LRF ($\log_{10}$)	7	0.714	0.428	0.364	2.5	0.236	0.122
Step yield	7	0.709	0.418	-2.23	2.43	0.242	0.0073

Table 5.3: Screening effect estimates for MVM LRF.

Term	Effect	Coef.	Std. err.	t	p-value	Sig.
A	-1.61	-0.807	0.2	-4.04	0.0273	*
B	-0.246	-0.123	0.2	-0.617	0.581	
AB	0.0192	0.00962	0.2	0.0482	0.965	

MVM LRF was impacted by the filtration volume. The effect is a loss of 1.61 log across the characterized range of the load, at $p = 0.027$. The filtration pressure returned an effect of 0.25 log at $p = 0.581$, and the interaction of the two parameters was smaller still. The screening model as a whole returns $p = 0.096$ against an R^2 of 0.848, which is the expected outcome of a design that spends 3 of its 6 degrees of freedom on terms that carry almost no effect.

No factor had a significant effect on XMuLV LRF or on step yield. The largest term for XMuLV LRF is the filtration pressure at $p = 0.120$, and the largest for step yield is the interaction at $p = 0.089$. The full effect tables for both responses are given in Appendix C.

The screening design identifies which factors act. It does not resolve curvature, and the effect magnitudes it returns are refined by the response-surface model reported next.

5.3 Response-surface models

Model adequacy for the three responses of the response-surface design is given in Table 5.4.

Table 5.4: Response-surface model adequacy for the three responses.

Response	N	R^2	Adj. R^2	Pred. R^2	F	p-value	RMSE
MVM LRF ($\log_{10}$)	12	0.947	0.903	0.527	21.4	0.000921	0.211
XMuLV LRF ($\log_{10}$)	12	0.506	0.0949	-1.14	1.23	0.398	0.158
Step yield	12	0.26	-0.356	-1.56	0.422	0.819	0.00685

The model for MVM LRF is adequate. It returns an R^2 of 0.947, an adjusted R^2 of 0.903 and an F of 21.4 at $p = 0.0009$. The predicted R^2 is 0.527, well below the adjusted R^2. A model whose predicted R^2 falls that far below its adjusted R^2 reproduces the runs it was fitted to better than it predicts a run withheld from it, which is a property of a design of 12 runs and a bound on how far the surface should be read. The partition of the residual is given in Table 5.5. Lack of fit returns $F = 5.55$ at $p = 0.096$, so the quadratic form is not rejected at $\alpha = 0.05$. The margin is not large, and the pure error it is tested against rests on 3 degrees of freedom.

Table 5.5: Analysis of variance for the MVM LRF response-surface model, with lack of fit tested against pure error.

Source	Sum sq.	df	Mean sq.	F	p-value
Model	4.75	5	0.951	21.4	0.000921
Residual	0.266	6	0.0444	nan	nan
Lack of fit	0.226	3	0.0752	5.55	0.0965
Pure error	0.0406	3	0.0135	nan	nan

The coefficients of the model are given in Table 5.6. The filtration volume carries the only significant term, at -0.866 log per coded unit and p = 5.6e-05. In natural units the model predicts 6.47 log at a load of 50 L/m2 and 4.71 log at 140 L/m2, both at the set-point pressure. The quadratic term in the filtration pressure is -0.301 at $p = 0.058$, which does not reach significance. It gives the surface a shallow maximum in pressure. Across the whole characterized pressure range at the set-point load the predicted MVM LRF moves from 5.36 log at 8 psi to 5.28 log at 30 psi, a range of less than half the effect of the load. The interaction of the two parameters is not significant.

Table 5.6: Full quadratic coefficients for MVM LRF on coded factors.

Term	Coef.	Std. err.	t	p-value	Sig.
Intercept	5.52	0.0962	57.4	1.87e-09	***
A	-0.866	0.086	-10.1	5.57e-05	***
B	-0.0341	0.086	-0.396	0.706	
AB	0.028	0.105	0.266	0.799	
A^2	0.136	0.129	1.05	0.333	
B^2	-0.301	0.129	-2.34	0.0582	

The surfaces of the three responses over the two parameters are shown in Figure 5.1. The MVM panel falls from left to right and is almost flat vertically, which is the graphical form of a model with one active factor. Diagnostics for the MVM model are shown in Figure 5.2. The residuals are unstructured against the predicted value, the normal plot is close to linear, and the actual against predicted plot shows no run driving the fit.

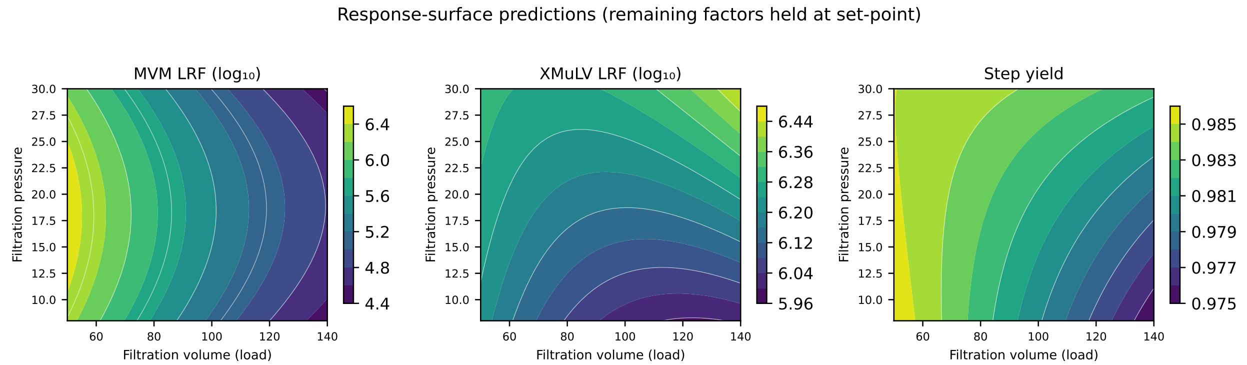

Figure 5.1: Response-surface predictions for MVM LRF, XMuLV LRF and step yield over the filtration volume and the filtration pressure.

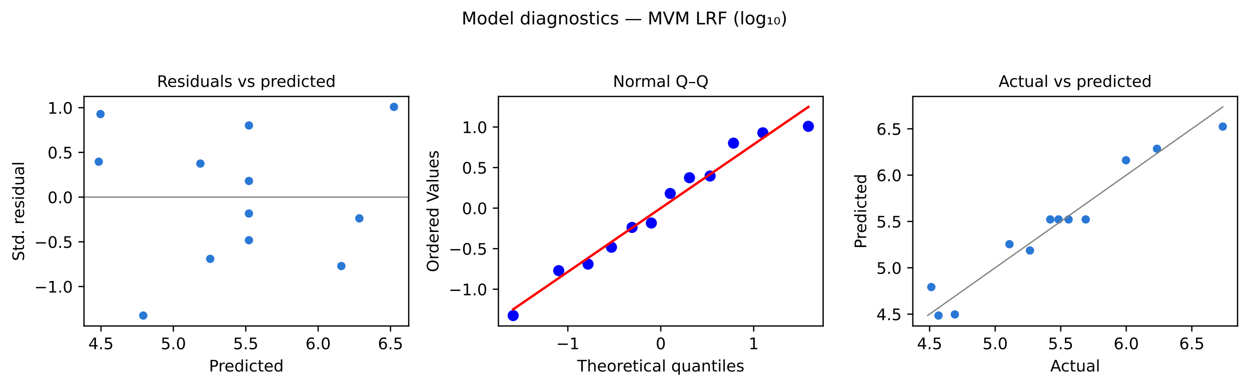

Figure 5.2: Model diagnostics for the MVM LRF response-surface model.

Neither XMuLV LRF nor step yield has an active factor. The XMuLV model returns F = 1.23 at p = 0.398, and no term in it reaches significance. Its lack of fit is not significant either, at p = 0.655, and most of its residual is the pure error of the centre points. The step yield model returns F = 0.42 at p = 0.819, with a centre-point mean of 98.3 %. Both responses are reported as robust over the characterized region rather than as fitted surfaces, because a model with no significant term predicts nothing that the centre-point mean does not. Their coefficient tables are given in Appendix C.

5.4 Mechanistic interpretation

The shape of the MVM surface follows from the sieving mechanism and from how the membrane fouls. The membrane retains virus by size, and the parvovirus is close to the pore size cut-off. As product and residual impurity accumulate on and within the membrane, the flux redistributes toward the least fouled and largest pores, which are the pores through which a particle of that size can pass. The further the batch is filtered, the larger the share of the flow that passes through them. Retention therefore falls as the load rises, and the load is the parameter that bounds the clearance claim.

The predicted loss of 1.76 log across the characterized load range is the size of that mechanism over a load range wider than the process will ever run.

XMuLV behaves as the same mechanism predicts at the other end of the size scale. The enveloped retrovirus is many times larger than the pores of the membrane, so it is retained on any pore that remains intact, and the redistribution of flux that lets a parvovirus through does not open a path for it. The model finds no active factor, and the lowest prediction anywhere in the characterized region is 6.01 log against a step criterion of 5.47 log. Retention of this virus is bounded by the integrity of the membrane rather than by the operating conditions, and integrity is confirmed on every batch by the post-use test under SOP-2012.

Step yield behaves the same way and for a related reason. The antibody is small enough to pass the pores freely, so the only loss is product held up in the fouled membrane and in the device. That loss is small and it does not vary appreciably with either parameter. The model predicts 98.2 % at the set-point and 97.7 % at the maximum load studied.

The pressure behaviour is the one part of the surface that the data support only weakly. The mechanism predicts a loss at both ends of the pressure range. At low pressure the flux is low and virus held within the pore structure has time to diffuse through the membrane, and at high pressure the fouling layer is compressed and the flux through the largest pores rises. A shallow maximum is what that pair of effects produces, and it is what the quadratic term in pressure describes. The term does not reach significance at $\alpha = 0.05$, so the pressure dependence is consistent with the mechanism rather than demonstrated by this study. It is treated as a bound on the pressure range in §7 and not as a predictive relationship.

6 Design space

The design space of this step is the region of the filtration volume and the filtration pressure over which both governed attributes meet the log reduction credited to the step. It is a subset of the characterized region, which is the knowledge space the two designs cover, and it contains the normal operating range within which the process is run.

The region was evaluated on an evenly spaced grid of 121 points across the characterized region, using the mean predictions of the two response-surface models. Of that grid, 75.2% meets both criteria. The response that rejects the remainder is MVM LRF ($\log_{10}$), which alone rejects 24.8% of the grid, while XMuLV LRF rejects none of it. The excluded part is the high load region, and its boundary runs almost vertically in Figure 5.1 because the load is the only active factor. The design space of this step is therefore bounded in one direction only.

The normal operating range sits inside the characterized region for both parameters, at 50 to 105 L/m2 for the filtration volume and 10 to 20 psi for the filtration pressure. At the upper edge of that range the model predicts 5.26 log of MVM clearance at the set-point pressure, which is 0.29 log above the step criterion.

Two bounds apply to the region as stated here. The grid is evaluated on mean predictions, so a point on the boundary meets the criterion with about even probability and not with assurance. The operating limits carried into Stage 2 are therefore taken from the proven acceptable ranges of §7, which require a 95 % predictive interval to sit inside the criterion and are narrower. The second bound is scale. The region was established on a small scale spiking model and its edges have not been run at commercial scale.

Working within the design space is not a change from a regulatory filing perspective (International Council for Harmonisation 2009). Planned movement within it is still assessed under the pharmaceutical quality system before it is made (International Council for Harmonisation 2008). Operation outside it is a deviation, is investigated under the quality system, and requires the clearance credited to the step to be reassessed for the batch concerned.

7 Proven acceptable ranges

7.1 Acceptance basis

The criterion this step is judged against is not the drug substance requirement. Both attributes it governs are cumulative over three orthogonal steps, and the requirement for MVM clearance is at least 8.6 log and for XMuLV clearance at least 16.7 log across the process. The criterion applied here is the log reduction credited to this step in the modular claim, less an allowance of 0.35 log for the variability of the infectivity assays. That gives at least 4.97 log for MVM and at least 5.47 log for XMuLV, and these are the criteria used in the analyses below and in the design space of §6.

The alternative basis is the backward calculation from the cumulative requirement. Subtracting the clearance credited to the low pH hold and to the anion exchange step from the cumulative requirement leaves 3.89 log for MVM and 3.65 log for XMuLV at this step. Both are lower than the criteria applied. The step is held to the claim made for it rather than to the residual requirement, because the claim is what the modular clearance argument in PCMR-001 adds up, and because a step that met only the residual requirement would leave no margin for the other two steps to lose.

7.2 Ranges

The proven acceptable ranges are given in Table 7.1. Each row is one attribute against one parameter, with the characterization range, the criterion and the two analyses described in §3.5. The first analysis fixes the other parameter. The second propagates the other parameter across its normal operating range by Monte-Carlo of the fitted model and requires the 95 % predictive interval to stay inside the criterion, which makes it the narrower of the two.

Table 7.1: Proven acceptable ranges per governed attribute and parameter.

CQA	Parameter	Char. range	Unit	Crite-rion	PAR (set-point)	PAR (NOR)
MVM LRF ($\log_{10}$)	Filtration volume (load)	50–140	L/m2	≥ 4.97	50–127	50–97.2
MVM LRF ($\log_{10}$)	Filtration pressure	8–30	psi	≥ 4.97	8–30	12.1–24.8
XMuLV LRF ($\log_{10}$)	Filtration volume (load)	50–140	L/m2	≥ 5.47	50–140	50–140

CQA	Parameter	Char. range	Unit	Crite-rion	PAR (set-point)	PAR (NOR)
XMuLV LRF ($\log_{10}$)	Filtration pressure	8–30	psi	≥ 5.47	8–30	8–30

The rows that carry the argument are the two MVM rows. Under the second analysis the proven acceptable range of the filtration volume ends at 97.2 L/m2, below the upper limit of the normal operating range of 105 L/m2. The routine maximum load is therefore not supported over the whole of the normal operating range once the pressure is allowed to vary within its own, and the maximum load carried into Stage 2 is the proven acceptable range and not the normal operating range (§10). The proven acceptable range of the filtration pressure under the same analysis is 12.1 to 24.8 psi. Its lower edge sits above the lower limit of the normal operating range of 10 psi and its upper edge sits above the upper limit of 20 psi, so the pressure range is bounded from below by this analysis and not from above.

The pressure bounds rest on the quadratic term reported in §5.3, which did not reach significance at $\alpha = 0.05$. They are therefore conservative rather than demonstrated, and they are treated as an operating limit rather than as evidence of an effect. The XMuLV rows return the whole characterized range under both analyses, which is the expected result for a virus retained on any intact pore.

Figure 7.1 shows the analysis for MVM LRF against the filtration volume, Figure 7.2 the same attribute against the filtration pressure, and Figure 7.3 the XMuLV attribute against the parameter with its largest main effect. In each figure the acceptance limit is drawn as a dashed line, the acceptable region of the parameter is shaded green, the set-point is marked and the normal operating range is shaded grey. The right panel of each figure carries the 95 % predictive interval of the second analysis.

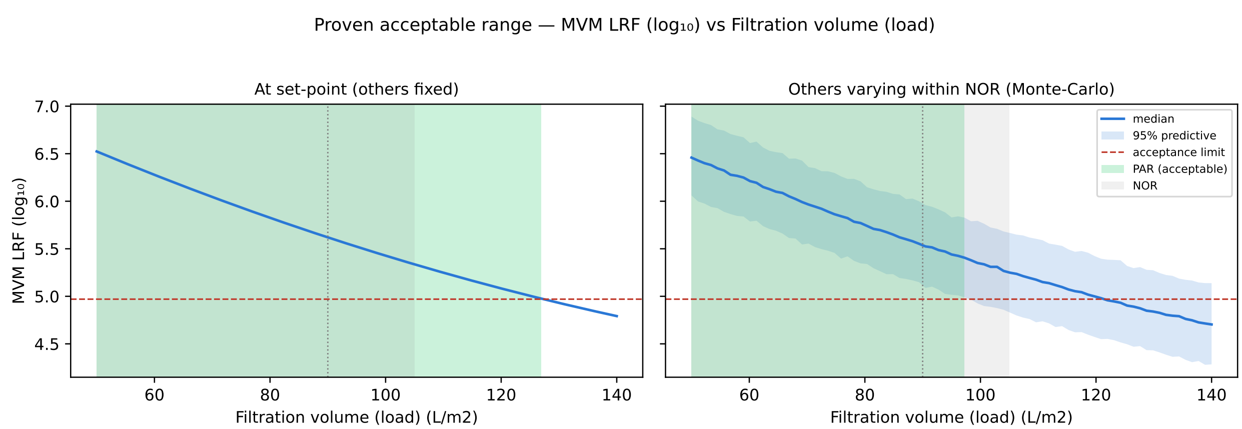

Figure 7.1: Proven acceptable range for MVM LRF against the filtration volume. The right panel propagates the filtration pressure across its normal operating range.

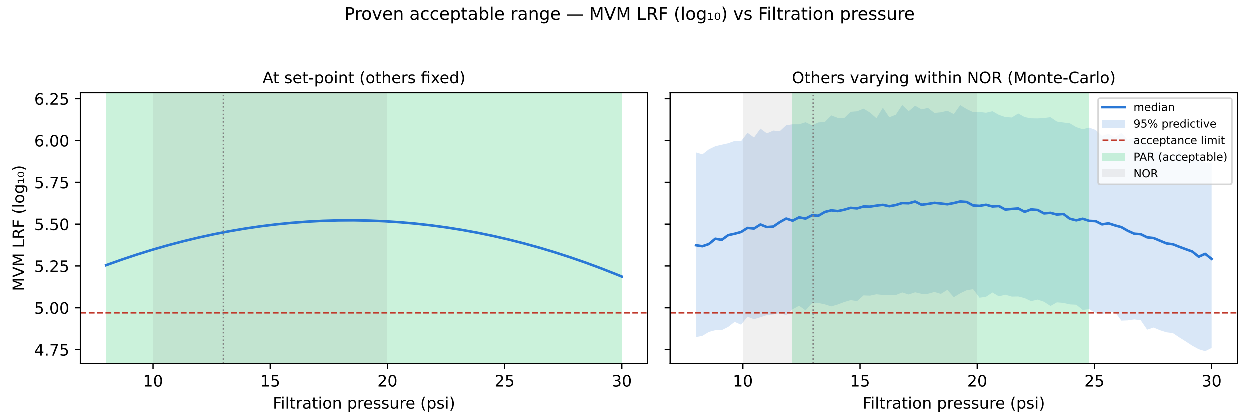

Figure 7.2: Proven acceptable range for MVM LRF against the filtration pressure. The right panel propagates the filtration volume across its normal operating range.

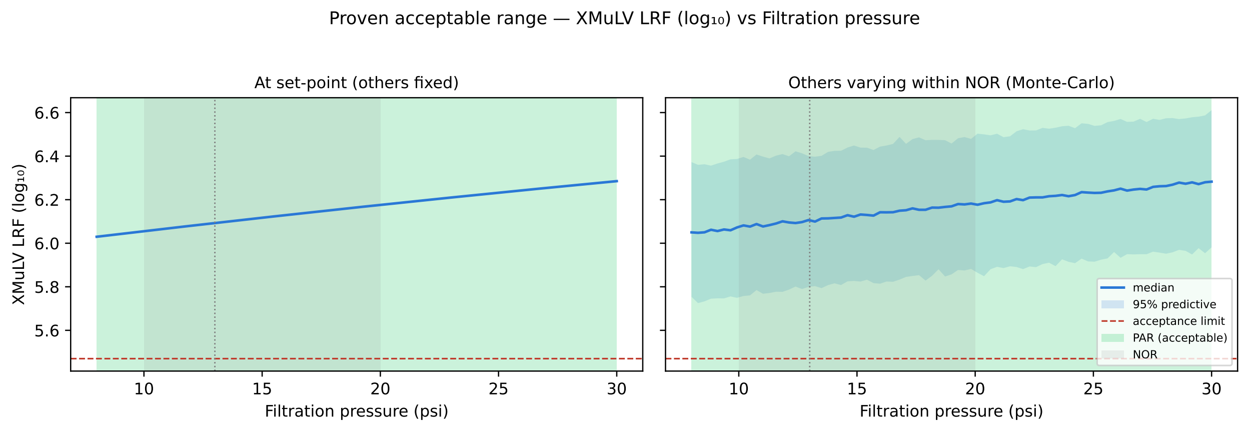

Figure 7.3: Proven acceptable range for XMuLV LRF against the filtration pressure, its largest main effect in the response-surface model.

The proven acceptable ranges are one dimensional sections through the region described in §6, and the multivariate region is the statement of record where the two disagree. A univariate range tells the reader how far one parameter may move. The design space tells them which combinations are acceptable. For this step the two are close, because only one parameter is active and the region is bounded in one direction.

8 Process capability and robustness

Capability at commercial scale was estimated by Monte-Carlo simulation of 2,000 batches of the whole process, with every parameter sampled inside its normal operating range and with the analytical and process variability of each step included. The two attributes this step contributes to are given in Table 8.1, with the mean, the standard deviation and the one-sided capability index against the lower limit.

Table 8.1: Commercial-scale capability of the two attributes the step governs.

CQA	Crit.	Spec	Acceptance	Mean ± SD	Cpk
Viral clearance — XMuLV (enveloped)	VH	lower	16.7–30	19.5 ± 0.61	1.53
Viral clearance — MVM (parvovirus)	VH	lower	8.6–20	10.4 ± 0.4	1.51

Cumulative MVM clearance is the tightest capability in the drug substance register, at a Cpk of 1.51. Its simulated mean is 10.38 log against a requirement of at least 8.6 log, and the standard deviation of 0.40 log is what brings the index down to that value. Cumulative XMuLV clearance carries a Cpk of 1.53 against a requirement of at least 16.7 log.

The capability of both attributes is a property of the process and not of this step. This step contributes 5.32 log of the 10.03 log of cumulative MVM clearance and 5.82 log of the 18.87 log of cumulative XMuLV clearance, and the balance comes from the low pH hold and the anion exchange step. It is the largest single contributor to the MVM claim, which is why the load bound of §7 matters to an attribute it does not set.

Robustness of the step itself is carried by the two responses with no active factor. XMuLV LRF and step yield were unchanged across a load range and a pressure range wider than the process will run, and the lowest XMuLV prediction anywhere in the characterized region is 6.01 log against a step criterion of 5.47 log. MVM LRF is the response that is not robust in this sense, and it is bounded by control of the load rather than by the width of its own variability.

Two bounds apply to these figures. They are estimated from qualified scale-down models and not from commercial batches, and they will be confirmed against the process performance qualification batches in Stage 2. The clearance figures are also modular, in that each step was spiked and measured independently, and the cumulative claim assumes the three mechanisms are orthogonal.

9 Parameter classification

Both parameters of the step were classified under SOP-4001 from the effects demonstrated in §5 and the ability of the commercial equipment to hold them. The classification is carried in Table 4.1. Neither parameter was classified as a critical process parameter, and no parameter of this step is a key or general process parameter.

- Filtration volume (load). Classified WC-CPP. The load has a demonstrated effect on MVM clearance, which is a very high criticality attribute, and it is the parameter that bounds the clearance claim. It is not classified CPP because it is directly controlled. The volume filtered per unit area is metered and the membrane area is fixed at installation, so the load reaches its limit only by a deliberate change to the batch record or by a metering failure that in-process monitoring detects. The proven acceptable range of 50 to 97.2 L/m2 contains the set-point of 90 L/m2, and its upper edge lies 7.25 L/m2 above that set-point.
- Filtration pressure. Classified WC-CPP. Its main effect on MVM clearance was not significant in either design, and its quality linkage rests on the shallow maximum described in §5.4 and on the bound it places on the range in §7.2. It is not classified CPP because the pressure is held by a pump control loop, is recorded continuously and alarms on excursion, which is what detected the deviation reported in §13.1. A parameter whose excursions are detected within the batch and whose effect within the characterized range is under half a log carries a low risk of taking a batch outside the design space.

The classification of this step contributes 2 well controlled critical process parameters to the register consolidated in PCMR-001. Neither is the single critical process parameter of the process, which is the pH of the low pH hold reported in PCR-006.

10 Contribution to the control strategy

The step contributes four elements to the control strategy of the drug substance process.

The first is the parameter ranges. The limits proposed for Stage 2 are the intersection of the normal operating range with the proven acceptable range of §7.2. For the filtration volume that gives an upper limit of 97.2 L/m2, which is tighter than the normal operating range. For the filtration pressure it gives 12.1 to 20 psi, which raises the lower limit of the routine range. Both limits are set from the analysis that propagates the other parameter, so a batch operated anywhere inside them is supported whatever the other parameter does within its own range.

The second is in-process monitoring. The filtration pressure is monitored continuously against its range and alarms on excursion. The volume filtered per unit area is recorded and is the value against which the batch is released from the step. The post-use integrity test under SOP-2012 is the control that confirms the membrane was intact for the batch, and it is the control on which the XMuLV claim rests, since no operating parameter bounds that attribute.

The third is control of the incoming material. The membrane lot is verified before use by the flux check described in §13.2, and the filter is used for one batch and discarded, so there is no life-cycle control of the kind a chromatography resin needs. The load material is the anion exchange flow-through pool, whose composition is controlled by the ranges reported in PCR-008.

The fourth is what this step does not control. No release test measures viral clearance, because clearance is claimed from the modular studies rather than from testing the drug substance (International Council for Harmonisation 2023a). No product quality attribute is controlled here. The glycan, charge variant and aggregate attributes are set at the bioreactor and reported in PCR-003, host cell protein and DNA are cleared by the chromatography steps reported in PCR-005, PCR-007 and PCR-008, and the drug substance concentration is set at the formulation step reported in PCR-010. The contribution of this step is consolidated with the rest of the train in PCMR-001.

11 Discussion

The study establishes what governs virus retention at this step and where the claim stops. One parameter acts. Retention of the parvovirus falls as the volume filtered per unit area rises, at 1.76 log across a load range wider than the process will run, and the response-surface model that describes it is adequate on every criterion applied to it. The enveloped retrovirus and the step yield are insensitive to both parameters across the same region. That pattern is what the sieving mechanism predicts, and it is the pattern reported for this filter type on three prior products.

Confidence in the scale-down model rests on three things. The model uses the same membrane and format as the commercial device and was operated at the same load per unit area and the same pressure. Its centre-point reproducibility is 2.10 %CV for MVM LRF, which is the scale of the assay rather than of the filtration. Its qualification compared input and output attributes against at-scale data under SOP-1001. The limit of that argument is that a spiking study cannot be run at commercial scale, so the clearance credited to the step is a model result throughout, as Q5A(R2) anticipates.

The deviations recorded at this step were dispositioned as retained. A transient pressure excursion reached 22 psi, inside the characterized range and inside the proven acceptable range for pressure, and a membrane lot was verified 12 % below the vendor typical flux before use. Neither changed the quantity the claim rests on, and both are set out in §13.

Three limitations bound what this report concludes. The design space of §6 was established on mean predictions of a design of 12 runs whose predicted R^2 is 0.527, so the surface should not be read far from the runs that support it. The pressure bounds of §7.2 rest on a term that did not reach significance, and they are applied as a conservative operating limit rather than as a demonstrated relationship. The edges of the region have not been run at commercial scale, and Stage 2 will confirm the step at its set-point and within the ranges proposed here rather than at those edges. Each of the three is managed by the same means, which is that the process is operated well inside the region and the load is bounded by the batch record.

The study also leaves one question to the master report. The cumulative clearance claim depends on the three viral steps being orthogonal, and this report can only show that this step's mechanism is size based and independent of the pH and charge mechanisms used by the other two. The orthogonality argument is made in PCMR-001, where all three contributions are consolidated.

12 Conclusions

The step was characterized over a region wider than the ranges it will be run in, and it meets the log reduction credited to it across that region except at loads above the bound established here. Filtration volume is the parameter that bounds MVM clearance, and its proven acceptable range under propagation of the filtration pressure ends at 97.2 L/m2. Filtration pressure has no significant effect on either attribute and is bounded from below at 12.1 psi. XMuLV clearance and step yield are robust across the whole characterized region.

Both parameters were classified as well controlled critical process parameters. The operating limits proposed for Stage 2 are the intersection of the normal operating range with the proven acceptable ranges, which tightens the maximum load and raises the minimum pressure relative to the ranges in PCP-009. Cumulative MVM clearance carries a Cpk of 1.51 and cumulative XMuLV clearance a Cpk of 1.53, both estimated from qualified scale-down models and both to be confirmed in Stage 2. The deviations recorded during execution were investigated and dispositioned as retained (§13). The step is ready for process performance qualification, and its outcome is consolidated in PCMR-001.

13 Deviations from the plan

During execution 2 deviations were recorded. Each was investigated and each was dispositioned as retained. The register is given in Table 13.1. Neither deviation invalidated a design, and the results reported in §5 use the complete dataset of both designs.

Table 13.1: Deviations recorded during execution of the study.

Deviation	Summary	Detected during	Disposition
DEV-009-01	Transient filtration-pressure excursion above NOR	in-process pressure monitoring	retained
DEV-009-02	Filter membrane lot flux below vendor-typical value	pre-use flux verification	retained

13.1 Transient filtration pressure excursion (DEV-009-01)

The filtration pressure reached 22 psi during one characterization run, above the upper limit of the normal operating range of 20 psi. The excursion was detected by the in-process pressure monitoring and was corrected within the run. The investigation traced it to overshoot of the pump ramp at the start of filtration, before the pressure control loop settled. The excursion was transient and the volume filtered per unit area was unaffected.

The impact was assessed against the pressure behaviour reported in §5. The pressure at which the excursion peaked lies inside the characterized range of 8 to 30 psi and inside the proven acceptable range of 12.1 to 24.8 psi established in §7.2. The response-surface model predicts 5.59 log of MVM clearance at that pressure and the set-point load, against a step criterion of 4.97 log. Pressure has no significant main effect on either governed attribute. The run was therefore retained and the data are included in the analysis.

The deviation is carried forward as evidence for the control it exercised. The pressure loop detected an excursion of this size within the batch, which is part of the reason the parameter is classified as well controlled in §9, and the continuous pressure record is named as an in-process control in §10.

13.2 Membrane lot flux below the vendor typical value (DEV-009-02)

The pre-use flux verification of membrane lot LOT-VF-9331 returned a flux 12 % below the value typical of the vendor's release data. The deficit was found before the lot was used, at the flux verification that SOP-2012 requires on every device. The investigation attributed it to permeability variation between membrane lots and found no fault in the device, the assembly or the buffer.

The impact was assessed against what the design controls. Both parameters of this step are set in terms the flux does not enter. The load is the volume filtered per unit area and the pressure is the transmembrane pressure applied across the membrane. A less permeable lot therefore filters the same load at the same pressure over a longer time. Retention depends on the load the membrane has seen and not on how quickly it saw it (§5.4). Lot to lot and device to device variation is also inside the pure error of both designs, because each centre-point replicate was executed on a separate filter device, and that pure error is the term reported in §5.1 and used in every lack-of-fit test. The lot was used, the runs executed on it were retained, and the responses were measured under AMV-3018 and AMV-3017 as for every other run.

The flux verification is retained as an incoming control on the membrane lot and is named in §10. A lot that fails it is not used, which is the control that made this a deviation with a bounded consequence rather than an unmeasured one.

14 Appendices

14.1 Appendix A. Screening design matrix and responses

The coded screening matrix and the responses measured on it are given in Table 14.1. Coded settings are minus one and plus one at the edges of the characterization range and zero at its midpoint. The factor letters are given in Table 4.2.

Table 14.1: Screening design matrix in coded factors, with the measured responses.

Run	Type	A	B	MVM LRF ($\log_{10}$)	XMuLV LRF ($\log_{10}$)	Step yield
1	factorial	-1	-1	6.34	6.33	0.985
2	factorial	-1	1	6.08	6.41	0.972
3	factorial	1	-1	4.71	6.06	0.972
4	factorial	1	1	4.48	6.51	0.996
5	center	0	0	5.33	6.46	0.985
6	center	0	0	5.01	6.17	0.994
7	center	0	0	5.97	6.28	0.979

14.2 Appendix B. Response-surface design matrix and responses

The coded response-surface matrix and the responses measured on it are given in Table 14.2. The axial runs are the rows with one factor at an edge and the other at zero.

Table 14.2: Response-surface design matrix in coded factors, with the measured responses.

Run	Type	A	B	MVM LRF ($\log_{10}$)	XMuLV LRF ($\log_{10}$)	Step yield
1	factorial	-1	-1	6.23	6.2	0.989
2	factorial	-1	1	6	6.34	0.985
3	factorial	1	-1	4.69	5.9	0.977
4	factorial	1	1	4.57	6.4	0.981
5	axial	-1	0	6.74	6.25	0.981
6	axial	1	0	4.51	6.38	0.979

Run	Type	A	B	MVM LRF (log_{10})	XMuLV LRF (log_{10})	Step yield
7	axial	0	-1	5.11	6.15	0.975
8	axial	0	1	5.27	6.28	0.985
9	center	0	0	5.48	5.94	0.99
10	center	0	0	5.56	6.29	0.989
11	center	0	0	5.69	6.03	0.972
12	center	0	0	5.42	6.27	0.982

14.3 Appendix C. Full effect and coefficient tables

The screening effects for the two responses not tabulated in §5.2 are given in Table 14.3 and Table 14.4, and the response-surface coefficients for the same two responses in Table 14.5 and Table 14.6. No term in any of the four tables reaches significance at $\alpha = 0.05$.

Table 14.3: Screening effect estimates for XMuLV LRF.

Term	Effect	Coef.	Std. err.	t	p-value	Sig.
B	0.262	0.131	0.0608	2.16	0.12	
AB	0.188	0.0939	0.0608	1.54	0.22	
A	-0.0821	-0.041	0.0608	-0.675	0.548	

Table 14.4: Screening effect estimates for step yield.

Term	Effect	Coef.	Std. err.	t	p-value	Sig.
AB	0.0181	0.00905	0.00365	2.48	0.0892	
A	0.00567	0.00283	0.00365	0.776	0.494	
B	0.00539	0.00269	0.00365	0.738	0.514	

Table 14.5: Response-surface coefficients for XMuLV LRF.

Term	Coef.	Std. err.	t	p-value	Sig.
Intercept	6.16	0.0719	85.7	1.7e-10	***
A	-0.0191	0.0643	-0.297	0.776	
B	0.128	0.0643	1.99	0.0943	
AB	0.0896	0.0788	1.14	0.299	
A^2	0.0849	0.0965	0.88	0.413	
B^2	-0.00707	0.0965	-0.0733	0.944	

Table 14.6: Response-surface coefficients for step yield.

Term	Coef.	Std. err.	t	p-value	Sig.
Intercept	0.982	0.00313	314	7.02e-14	***
A	-0.00336	0.0028	-1.2	0.275	
B	0.0016	0.0028	0.573	0.587	
AB	0.00199	0.00342	0.58	0.583	
A^2	-0.000294	0.00419	-0.0701	0.946	
B^2	0.000228	0.00419	0.0544	0.958	

14.4 Appendix D. Analytical methods summary

The validated methods used to measure the responses of this study are given in Table 14.7. Both are infectivity assays and both report a titre, from which the log reduction factor is calculated as the difference between the load and the filtrate on a common volumetric basis. Step yield was determined by mass balance from protein concentration and volume and carries no separate method validation.

Table 14.7: Validated analytical methods for the two model viruses.

Reference	Analyte	Technique	Reportable value
AMV-3018	MVM (minute virus of mice)	Infectivity titre (TCID50) with qPCR confirmation	Log reduction factor ($\log_{10}$)
AMV-3017	XMuLV (xenotropic murine leukaemia virus)	Infectivity titre (TCID50)	Log reduction factor ($\log_{10}$)

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